PROJECT REPORT

Google Cloud Network Intelligence Center Connectivity Troubleshooting

Cloud Computing • Virtual Networking • Firewall Analysis

Project Domain	Cloud Computing and Network Security
Platform	Google Cloud Platform (GCP)
Services Used	Compute Engine, VPC Network, Firewall Rules, Connectivity Tests

Prepared using the Google Cloud Console screenshots supplied for this project.

Abstract

This project demonstrates how Google Cloud Network Intelligence Center Connectivity Tests can be used to analyze communication between two virtual machine (VM) instances. A source VM and a destination VM were configured in a custom Virtual Private Cloud (VPC) network. A TCP connectivity test targeting port 80 was run to observe the expected path and connectivity status. A temporary firewall-rule scenario was then used to demonstrate how a firewall-related issue can make the destination unreachable. After the test rule was removed, the connectivity test was run again and reported successful delivery of all 50 packets. The exercise illustrates the value of connectivity testing for diagnosing network reachability and validating changes.

1. Introduction

Cloud applications rely on correctly configured networks, routes, virtual machine interfaces, and firewall rules. When a connection fails, manually inspecting each component can be time-consuming. Google Cloud Connectivity Tests provides configuration analysis and, where available, live data plane analysis to help identify whether traffic can travel from a source to a destination and which network component may affect that path.

In this lab, the test path was configured from **source-vm** to **destination-vm** over TCP port **80**. The screenshots show a successful initial test, a blocked test result, and a successful retest.

2. Objectives

• Create and inspect a custom VPC network and virtual machine instances.

• Configure an ingress firewall rule for TCP port 80.

• Use Connectivity Tests to examine the path from a source VM to a destination VM.

• Observe how a firewall-related rule can affect connectivity.

• Remove the test condition and confirm connectivity is restored.

• Capture evidence and document the configuration, test results, and lessons learned.

3. Tools and Technologies

Tool / Service	Purpose
Google Cloud Console	Web interface for creating and managing cloud resources.
Compute Engine	Hosts the source and destination VM instances.
Virtual Private Cloud (VPC)	Provides the network environment and subnet for the VMs.
VPC Firewall Rules	Controls permitted or denied network traffic.
Network Intelligence Center – Connectivity Tests	Analyzes the network path and reports reachability results.
TCP port 80	Port used as the connectivity-test destination port.

4. Network Configuration

The following names and values are visible in the supplied screenshots or were recorded during the lab.

Parameter	Value
VPC network	nic-troubleshooting-vpc
Subnet	nic-subnet (CIDR recorded during setup: 10.10.0.0/24)
Source VM	source-vm
Destination VM	destination-vm
Region / zone shown	asia-south1 / asia-south1-a
Test name	test-source-to-destination
Protocol / destination port	TCP / 80
Ingress allow rule	allow-web-server; target tag web-server; TCP port 80; priority 1000
Temporary test rule	deny-web-server-test; target tag web-server; TCP port 80; priority 900 (see evidence note in Section 7)

5. Implementation Procedure

Step 1 — Create the network and VM instances

A custom VPC network named **nic-troubleshooting-vpc** and a subnet named **nic-subnet** were used. Two Compute Engine instances, **source-vm** and **destination-vm**, were configured in the same network and zone shown in the screenshots. The destination VM was intended to receive test traffic on TCP port 80.

Step 2 — Configure the ingress firewall rule

The firewall rule **allow-web-server** was created for the **web-server** target tag and TCP port 80, with priority 1000. This rule was part of the successful connectivity path shown in the Connectivity Test trace.

Step 3 — Run the initial Connectivity Test

A Connectivity Test named **test-source-to-destination** was configured with source **source-vm**, destination **destination-vm**, protocol TCP, and port 80. The initial result showed both directions as reachable in configuration analysis and **50/50 packets delivered** in live data plane analysis.

Step 4 — Simulate a firewall-related failure

A temporary test rule named **deny-web-server-test** was introduced with priority 900 and associated with the **web-server** target tag. The subsequent Connectivity Test reported the forward path as **Unreachable** and live data plane analysis showed **0/50 packets delivered**. The trace identified the temporary rule as the point where packets could be dropped.

Step 5 — Restore and verify connectivity

After the temporary test rule was removed, the Connectivity Test was run again. The final screenshot shows configuration analysis as reachable in both directions and live data plane analysis reporting **50/50 packets delivered**. This confirmed that the test path was reachable again at the time of the retest.

6. Results and Screenshot Evidence

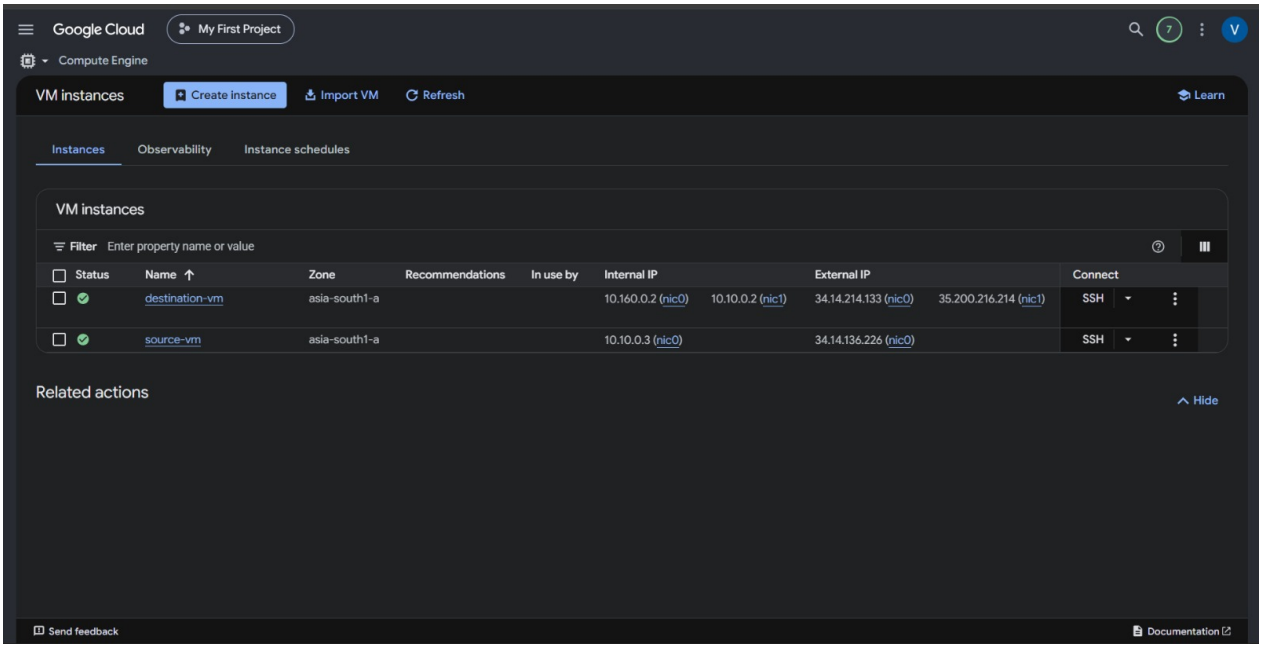


Figure 1. Compute Engine VM instances: source-vm and destination-vm.

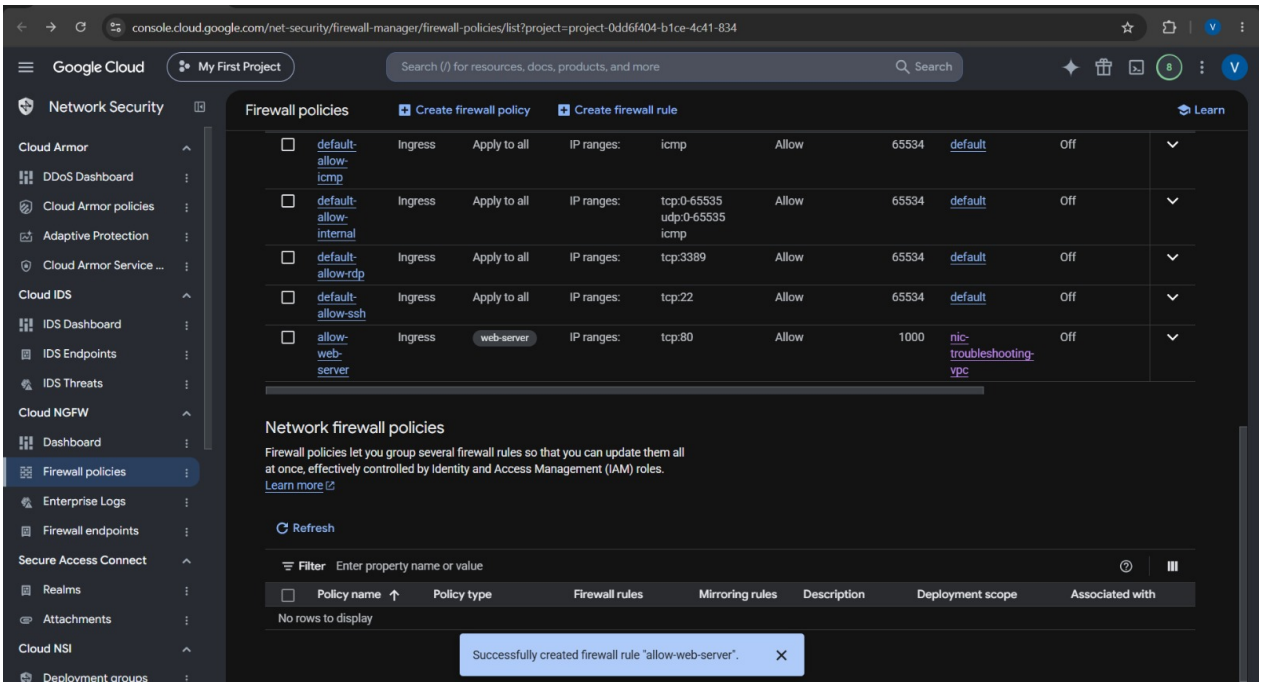


Figure 2. Ingress firewall rule allow-web-server for TCP port 80, priority 1000.

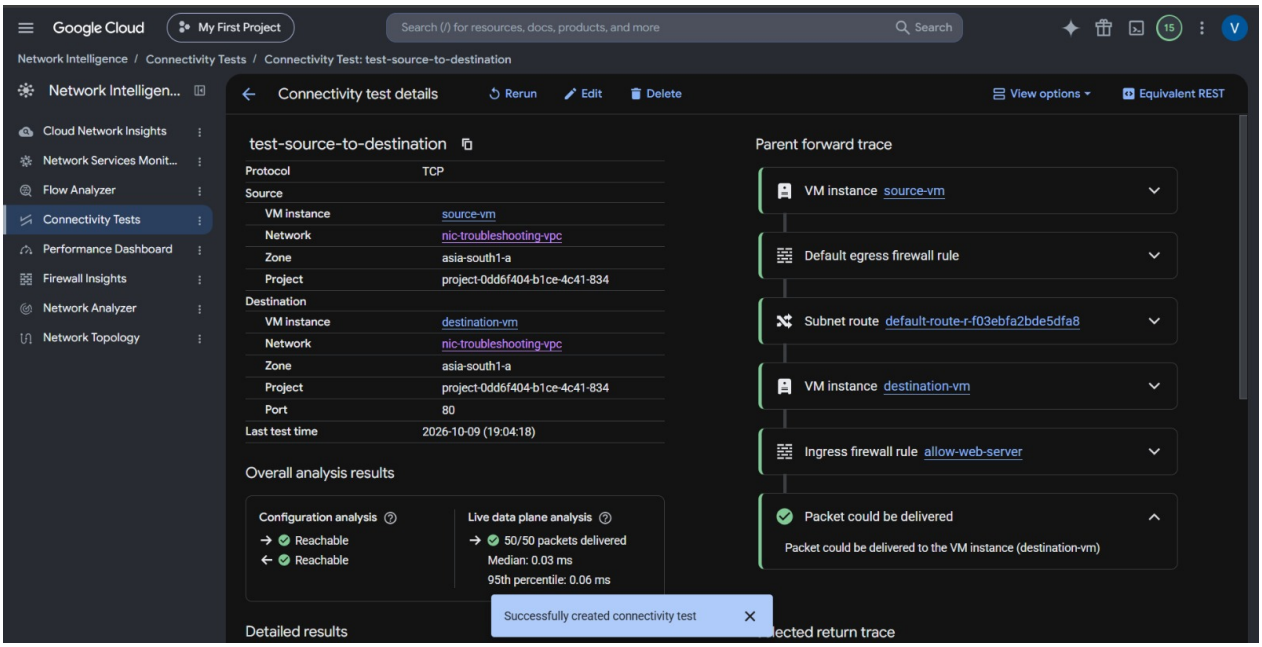


Figure 3. Successful Connectivity Test: both directions reachable and 50/50 packets delivered.

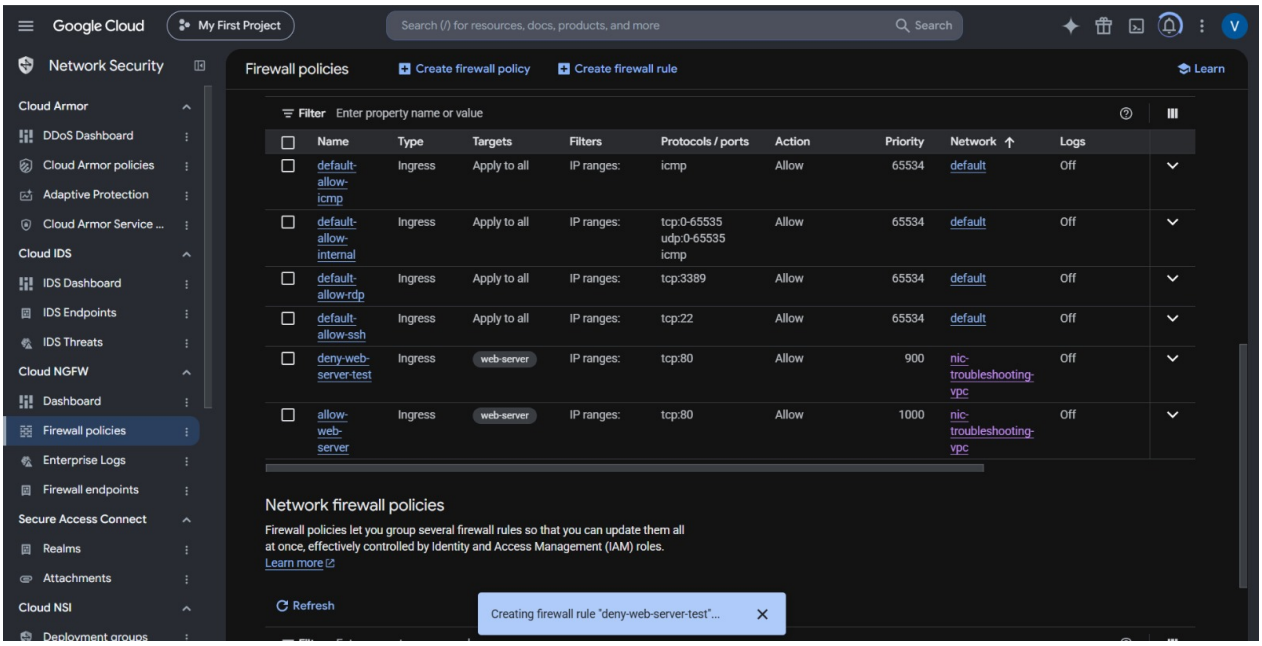


Figure 4. Firewall rule list during the failure simulation.

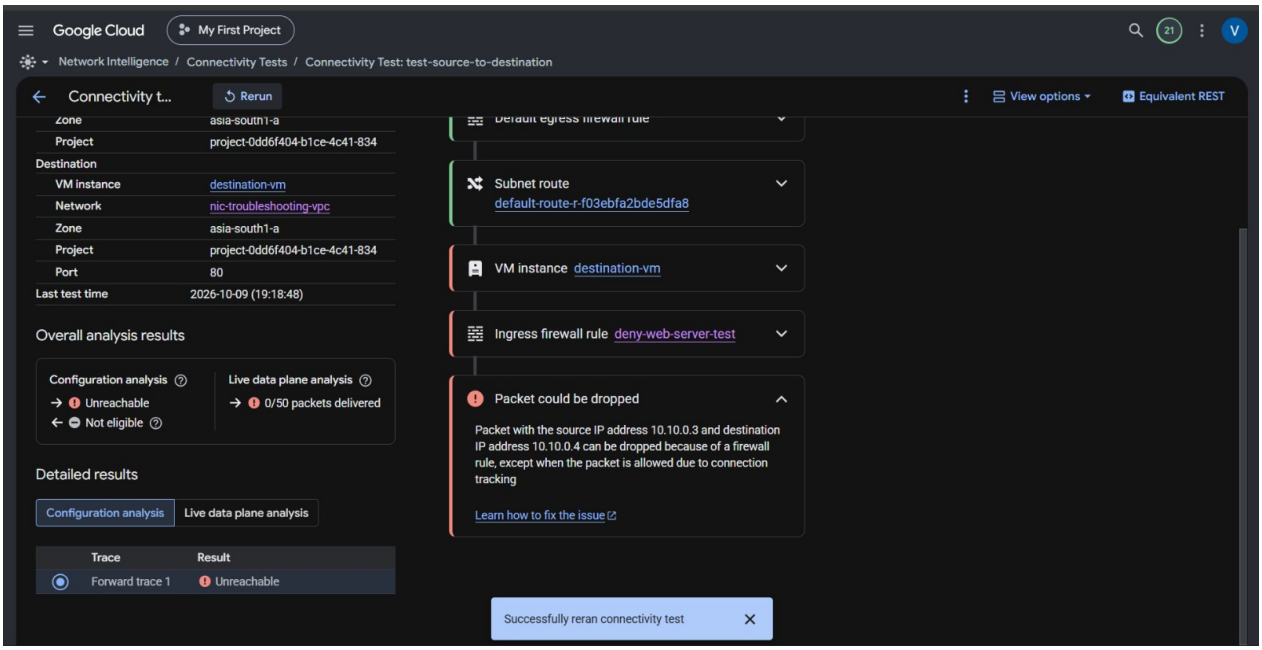


Figure 5. Connectivity Test reports Unreachable and 0/50 packets delivered; trace identifies deny-web-server-test.

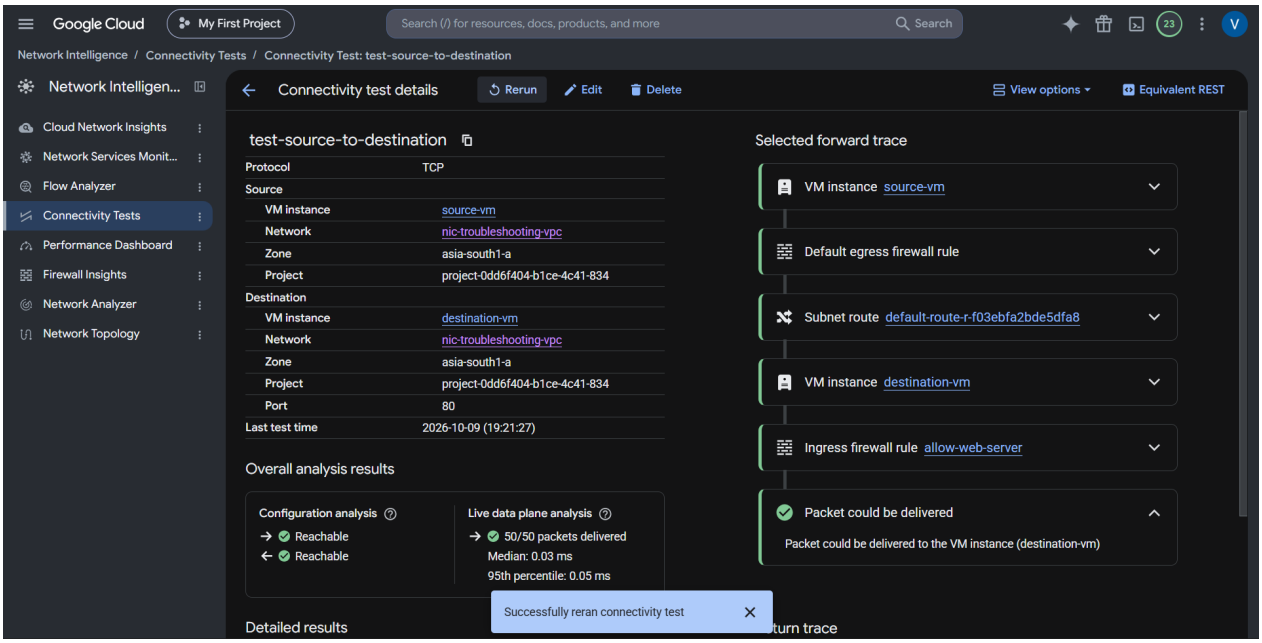


Figure 6. Connectivity restored: both directions reachable and 50/50 packets delivered.

7. Analysis and Discussion

The successful test demonstrates that Connectivity Tests could trace the selected TCP path from source-vm to destination-vm and report packet delivery. During the failure simulation, the test changed to Unreachable and the live data plane result fell to 0/50 packets delivered. The trace highlighted a firewall-rule stage, making the rule a useful starting point for troubleshooting. After the temporary test condition was removed, the retest returned to 50/50 packet delivery.

Evidence consistency note: In the supplied firewall-rule-list screenshot (Figure 4), the Action column for the row named *deny-web-server-test* appears to display *Allow*, while the Connectivity Test failure screenshot (Figure 5) identifies that same rule as the packet-drop point. The report preserves both observations rather than assuming the table screenshot is correct. If your instructor checks the rule configuration closely, capture a fresh screenshot of the rule details showing its actual action, or omit Figure 4 and rely on the Connectivity Test result screenshot.

8. Key Learnings

• Connectivity Tests can provide both configuration analysis and live data plane analysis.

• Firewall rules, including their action, priority, target tags, and port filters, are important when diagnosing connectivity.

• A test result is stronger when the configuration trace is reviewed together with packet-delivery evidence.

• After making a network change, rerunning the same test helps confirm whether the path has been restored.

• Cloud resources should be removed after a lab when they are no longer needed to avoid unintended ongoing usage.

9. Resource Cleanup

After collecting the screenshots, the lab resources were reported as deleted, including the VM instances, Connectivity Test, firewall rules created for the exercise, and the custom VPC/subnet. This cleanup reduces the risk of leaving resources active after the practical exercise.

10. Conclusion

This project used Google Cloud Network Intelligence Center Connectivity Tests to investigate reachability between two Compute Engine VM instances over TCP port 80. The screenshots document a successful baseline test, a failure state reporting 0/50 packets delivered, and a final successful retest reporting 50/50 packets delivered. The exercise demonstrates a practical troubleshooting workflow: establish a baseline, examine the network trace, isolate a possible firewall-related issue, apply a corrective change, and rerun the test to verify the outcome.

11. References

1. Google Cloud — Connectivity Tests documentation: https://cloud.google.com/network-intelligence-center/docs/connectivity-tests

2. Google Cloud — VPC firewall rules documentation: https://cloud.google.com/firewall/docs/firewalls

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